Mutation Research - Fundamental and Molecular Mechanisms of Mutagenesis 824 (2022) 111776

Contents lists available at ScienceDirect

Mutation Research - Fundamental and Molecular Mechanisms of Mutagenesis

journal homepage: www.elsevier.com/locate/mut

Cell cycle involvement in cancer therapy; WEE1 kinase, a potential target as therapeutic strategy

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ARTICLE INFO

Keywords:
Cell cycle
WEE1 kinase
Cancer
DNA damage
Checkpoint

ABSTRACT

Mitosis is the process of cell division and is regulated by checkpoints in the cell cycle. G1-S, S, and G2-M are the three main checkpoints that prevent initiation of the next phase of the cell cycle phase until previous phase has completed. DNA damage leads to activation of the G2-M checkpoint, which can trigger a downstream DNA damage response (DDR) pathway to induce cell cycle arrest while the damage is repaired. If the DNA damage cannot be repaired, the replication stress response (RSR) pathway finally leads to cell death by apoptosis, in this case called mitotic catastrophe. Many cancer treatments (chemotherapy and radiotherapy) cause DNA damages based on SSBs (single strand breaks) or DSBs (double strand breaks), which cause cell death through mitotic catastrophe. However, damaged cells can activate WEE1 kinase (as a part of the DDR and RSR pathways), which prevents apoptosis and cell death by inducing cell cycle arrest at G2 phase. Therefore, inhibition of WEE1 kinase could sensitize cancer cells to chemotherapeutic drugs. This review focuses on the role of WEE1 kinase (as a biological macromolecule which has a molecular mass of 96 kDa) in the cell cycle, and its interactions with other regulatory pathways. In addition, we discuss the potential of WEE1 inhibition as a new therapeutic approach in the treatment of various cancers, such as melanoma, breast cancer, pancreatic cancer, cervical cancer, etc.

1. Introduction

The decision about when a cell should divide or grow is very important in the cell life cycle. Cell cycle regulation is a very important process in the timing of this decision [1]. One hallmark of cancer development is sustained proliferation, caused by a lack or defect in the cell cycle regulation system [2]. The cell cycle consists of two major stages, S and M. In the S stage the chromosomes of the mother cell are duplicated into two identical chromatids. In the M stage the two identical chromatids are divided into each of the daughter cells [3]. The goal of the cell cycle regulation is to accurately transfer the genetic information to the next generation of cells. The regulation of the timing of these two stages is crucial, because if the M stage occurs before the chromosome replication is complete, at least one of the daughter cells will lack all the genetic informative [4]. The role of the checkpoints is to prevent the start of the next phase of the cell cycle before the completion of the previous phase [5]. Checkpoints include sensor systems that detect phase completion, as well as sensors that activate effector

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https://doi.org/10.1016/j.mrfmmm.2022.111776
Received 25 November 2021; Received in revised form 7 February 2022; Accepted 8 February 2022
Available online 19 February 2022
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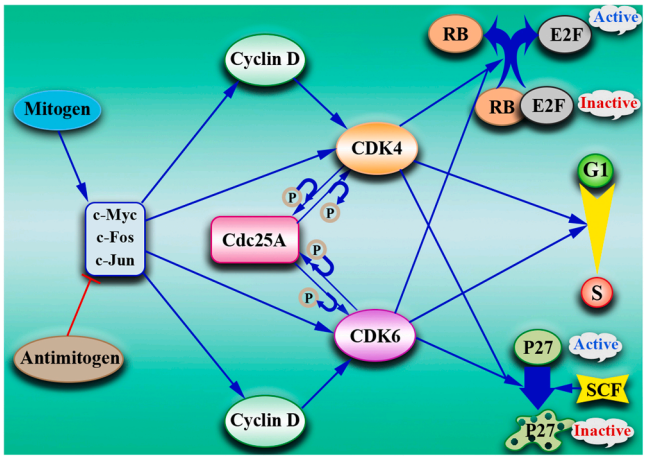


Fig. 1. G1-S transition regulatory mechanism. During the G1 phase, mitogen-induced signals lead to the activation of transcription factors such as C-Myc, C-Fus, and C-Jun, leading to the activation of cellular proliferation and mitosis. These transcription factors can activate cellular proliferation and mitosis by regulating the CDK4-cyclin D and the CDK6-cyclin D Complexes. These factors induce transcription of the cyclin D gene, Thereby cyclin D assembling with its partner CDK 4/6 and accumulates cyclin D – CDK 4/6 complexes levels in the nucleus. These complexes are direct regulators of the G1 to S progression, and can also act indirectly by removing the inhibitory effect of retinoblastoma (RB) protein on E2F, resulting in S phase progression. E2F is a transcription factor which primarily binds to and inhibits RB, but when it is detached from RB, it enhances cyclin A transcription and allows the cancer cells to enter the S phase. CDK 4/6, also in a negative feedback loop, inhibits p27 activation via inducing its degradation in the SCF ubiquitin ligase complex. When p27 binds to CDK4, 6-cyclin D complexes it leads to separation of CDKs from cyclins. Phosphorylation of p27 at T187 residue facilitates its degradation by the SCF ubiquitin ligase complex. Cyclin D- CDK 4/6 complex could also activate themselves in a positive feedback loop by activating CDC25 A as its activator. All these processes eventually lead G1 to S phase progression.

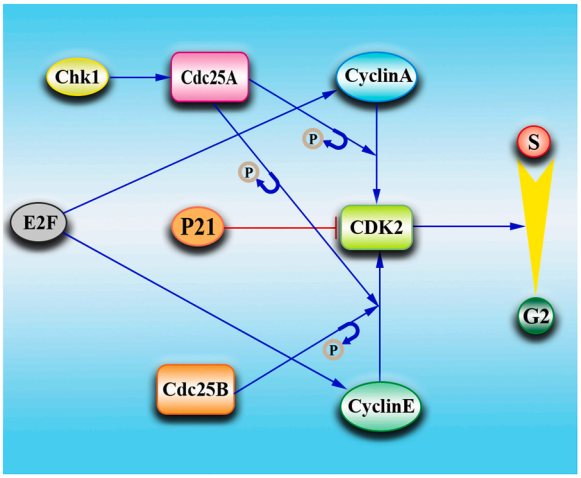


Fig. 2. S-G2 transition regulatory mechanism. During the S phase activated E2F in G1 induces cyclin A and cyclin E accumulation, which increases CDK 2 assembly with these cyclins (CDK2-cyclin A, E complexes are regulators of the S-phase progression). This process could be potentiated by the CDC25 A and B activatory effect to overcome p21 inhibitory effect on CDK 2 – cyclin A and cyclin E complexes. p21 is a cyclin dependent kinase inhibitor which inhibits CDK 1, 2- cyclin A, E activity. The CDC25 phosphatase enzymes (A, B, and C) are key components of cell cycle regulation by removing phosphate groups from CDK1, 2, 4, 6 and allowing complex formation with cyclins A, B, D, E. All these processes eventually lead to S-G2 transition.

pathways, which can arrest the cell cycle and then activate repair pathways [6]. Checkpoints are responsible for the high accuracy of cell division by monitoring growth, DNA replication, DNA damage, and kinetochore binding to the mitotic spindle [7]. Checkpoints are executed through cyclins, CDKs (cyclin-dependent kinase), CKIs (cyclin-dependent kinase inhibitor), and APC/C (anaphase-promoting complex/-cyclosome) [8]. The mechanisms of the different phase checkpoints are similar, in that the sensors detect disturbed cell growth, incomplete replication, DNA damage, and spindle defects, and then activate the signaling pathway response to activate the effectors. The effectors in turn trigger cell cycle arrest and activate the appropriate repair pathways [9–11]. The major cell cycle regulators are heterodimeric protein complexes, which have a kinase activity. The subunits of the heterodimers are comprised of a regulatory unit (cyclin) and a catalytic unit (CDK) [12]. The CDKs bind to different cyclins, thereby regulating the timing and initiation of the cell cycle phases. These complexes cause the phosphorylation of large number of proteins involved in cell cycle regulation, which can lead to activation or inhibition of each particular protein (Figs. 1, 2 and 3) [13].

These checkpoints, as previously mentioned, play an important role in cancer. Inactivation as well as increases in the normal activity of the checkpoints may lead to abnormalities in the replication process, such as genetic mutations or changes in the expression of some genes [14,15].

In this review, we look at how the normal cell cycle is performed and how it is regulated by CDK-cyclin complexes (which consist of cyclin regulatory and CDK catalytic subunits). We also explain how CDKs are regulated by CKIs and CAK (CDK-activating kinase), and how this regulation affects the cell cycle. Mitosis CDK (CDK1) is a key contributor to the initiation of mitosis by controlling proper replication, and the repair of DNA damage caused by inappropriate replication. Furthermore, we discuss the role of WEE1, a major CDK1 inhibitor protein that has recently been proposed to be a therapeutic target in many cancers.

2. CDK activity regulation by cyclins

So far there have been thirteen different CDKs discovered. Of these, CDK1, CDK 2, CDK 4 and CDK 6 are responsible for regulating cell cycle events (Figs. 1, 2 and 3) [16]. CDK4 and CDK6 trigger cells at the G1 phase to enter the cell cycle by binding to cyclin D [17]. CDK2 is responsible for the initiation of the S and G1-S phases by binding to cyclins A and E [18]. CDK1 determines the entry of dividing cells into the M phase by binding to cyclins A and B [19]. CDKs are not active in a monomeric form, and require the subunits to be activated together. Indeed the CDK catalytic subunit lacks any kinase activity until it is associated with the regulatory subunit [20]. It is noteworthy that each CDK can only bind to a limited number of cyclins in each phase of the cell cycle. Thereby, cyclins exert their regulatory role by being present only at the appropriate cell cycle phase and triggering the activation of the relevant CDKs [21,22]. Thus CDKs are activated during specific phases, leading to the progression from that particular stage to the next stage, and remain inactive at other stages [23–25]. This regulation is triggered by positive feedback mechanisms that cause each CDK to enhance its own activity, and these mechanisms can combine with a negative feedback mechanism, which causes each CDK to terminate its own activity [26,27]. One of the regulatory mechanisms for CDK activity is the transcriptional control of the regulatory subunits (cyclins) [28]. Indeed, transcriptional changes occurring in genes during the cell cycle leads to changes in the expression levels of specific cyclins during different phases [29,30]. For example, transcription of cyclins E and D is induced by the E2F transcription factor [31]. Surprisingly, there are also transcription factors genes for cyclins A and B among the genes that the E2F transcription factor causes to be expressed [30].

3. Other mechanisms for regulation of CDK activity

There are several mechanisms for regulating the activity of CDKs, so

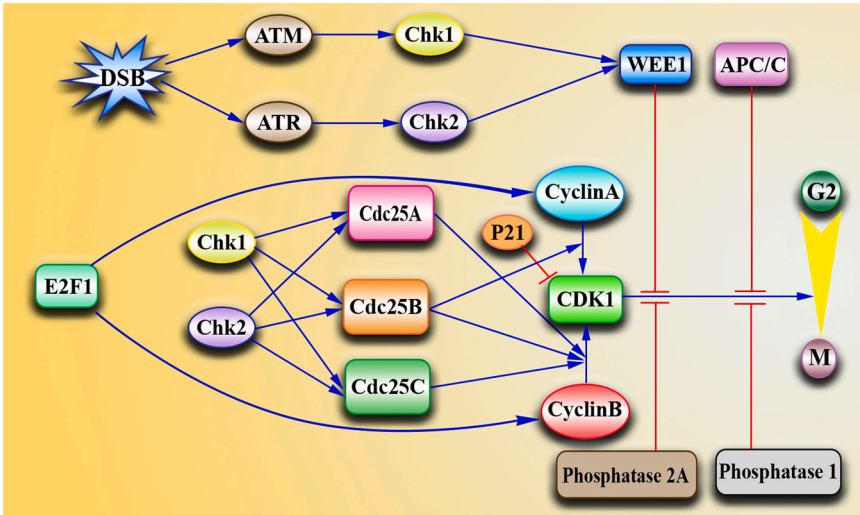


Fig. 3. G2-M transition regulatory mechanism. In G2-M transition, cyclin A and cyclin B- CDK 1 complex assembly due to the enhanced cyclin A and B transcription by the E2F leads to G2-M transition. APC / C, Phosphatase 1 and Phosphatase 2A also function normally and halt G2-M transition, thus preventing the two proliferated cells from dividing before the mitosis process is complete. Also, G2-M phase arrest happens when cells are exposed to DNA breaks, which will activate CHK1 and CHK2 by activating ATR and ATM pathways. In other words, DNA breaks activate ATR and ATM pathways, which in turn phosphorylate and activate both CHK1 and CHK2. Then, WEE1 kinase is activate in response to CHK1/2 activation, and regulates G2-M progression by inhibiting the CDK1-cyclinA, B complexes. It means activated WEE1 inhibits CDK1-cyclin B and delays the G2-M transition to provide an opportunity to repair DNA damages.

that these CDKs can be activated or inactivated at just the right time. CDK regulators are divided into three groups of kinases and phosphatases, inhibitory proteins, and protein ligases [1, 32–35]. Among the kinases, CAK (CDK activator kinase) activates the CDK-cyclin complex, but WEE1 kinase inhibits the CDK1-cyclin A/B complex [18]. Among the phosphatases, CDC25B, CDC25C, and CDC25A activate their targets depending on their particular substrate (CDK1/2/4/6-cyclin complexes) (Figs. 1, 2 and 3) [17]. CDC4 phosphatase inactivates CHK1 (checkpoint kinase 1) [1]. Among the inhibitory proteins, sic1 protein inhibits CDK2. The cyclin-dependent kinase inhibitor proteins: p22 kip1, p57 kip2, and p21 cip can inhibit all the CDKs. Ink 4 protein can inhibit CDK4, CDK6, and retinoblastoma protein [33]. Retinoblastoma binds to and inhibits the E2F transcription factor, to prevent the transcription of various genes related to the cell cycle (Fig. 1) [34]. SCF ligase protein, which derives its name from its Cullin, F-box, and Skp1 components, can activate CDK2 by degradation of Sic1 and p27 kip1 (Fig. 1). The APC/C-CDC20 protein ligase can degrade securin, which is a protein that controls the metaphase-anaphase transition and anaphase initiation. This degradation liberates separase, leading to chromosome separation. The APC/C-CDH1 complex inhibits CDK1 by inhibiting and degrading cyclin B (Fig. 3) [35].

The ubiquitination process is one mechanism to regulate cyclin levels, ensuring that the cell cycle is irreversible, because the ubiquitinated proteins are degraded in the proteasome complex [19]. In other words, when a particular cyclin degraded, the processes that are activated by that cyclin can no longer occur. Protein ubiquitin ligases prepare cyclins for proteasomal degradation by labeling them with ubiquitin [35]. Moreover, SCF can also lead to destruction of cyclins. SCF controls the passage of the cell cycle through the G1-S phase by degrading cyclins A, D, E, and CDK inhibitors (CKIs) [36]. APC/C can also degrade cyclins A, B, E, leading to the cell exit from mitosis [37]. The regulatory mechanism is different for these two ligases (SCF and APC/C). SCF is active throughout the cell cycle and recognizes phosphorylated substrates, therefore its activity is regulated by the phosphorylation of its substrate. In the regulation process of APC/C, the substrate is active throughout the cell cycle, and it is the activation of APC/C that regulates degradation process. At the initiation of anaphase, APC/C is activated by CDK1 and other protein kinases, which leads to the degradation of cyclins and other regulators. This process only occurs during the M and G1 phases [38,39].

CDKs are not only regulated by binding to cyclins, but can also be inhibited or activated by phosphorylation. Thus promoting or inhibiting the phosphorylation of subunits acts as a regulatory mechanism. By visualizing the spatial structure of CDKs it was discovered that when the CDKs were non-phosphorylated, they had a flexible region called the T-loop [40]. This loop is highly conserved and inhibits the kinase activity due to spatial blocking of the ATP binding site (active site). When a cyclin binds to CDK, the T-loop interacts with the cyclin making the CDK active site accessible, but nevertheless the CDK kinase activity remains low [41]. The activity increases after the phosphorylation of threonine in the T-loop region. This phosphorylation is mediated by CAK, and increases the substrate affinity by modifying the conformation leading to activation of CDK [42]. The level of CDK determines the extent of phosphorylation [43]. When the complex between CDK-cyclin is formed, the threonine is phosphorylated by CAK to activate the complex [44]. Inhibitory phosphorylation reactions at amino acids, tyrosine Y15 or threonine T14 also plays an important role in regulating the CDK1-cyclinA/B complex activity [43]. These two amino acids are located in the ATP binding cleft, and their phosphorylation prevents ATP from entering the active site. As a result, the activity of the complex is inhibited [43]. This phosphorylation, which plays a critical role in regulating CDK1 activity and cell cycle, is mediated by the highly conserved WEE1 kinase protein [42]. Also dephosphorylation of these amino acids is performed by the highly conserved CDC25C phosphatase protein, which results in CDK1 activation (Fig. 3) [45].

CKIs can bind directly to the CDK-cyclin complex to inhibit its activity [46]. These CKIs have little resemblance to each other, but their functions are similar in the inhibition of CDK activity [47]. The CKIs that regulate CDK1 and CDK2 activity include, p27 kip1, p21 cip, and p57 kip1 [48]. There are also a group of CKIs that can interact with CDK4 and CDK6, which are called INK4 (inhibitor of CDK4). INK4 inhibits CDK4 and CDK6 by preventing the formation of a complex with cyclin D [49]. It has been shown that DNA replication does not begin until these CKIs have been degraded [50].

4. Molecular events in cell cycle regulation

Mitogenic factors or growth inducers initiate the early stages of the cell cycle, by triggering a cascade of signaling pathways (Fig. 1) [51]. Growth inhibitors (anti-mitogenic factors) monitor growth and regulate cell division. Taken together, the mitogen triggers a tyrosine kinase receptor-dependent signaling pathway that then activates a signaling cascade prompting cell division [52]. These pathways can work in different ways; firstly, mitogens can activate transcription factors that initiate transcription of genes; secondly, mitogens can inhibit the transcription of CDK inhibitor genes thereby initiating cell division [53]. The genes that increase in response to mitogens can be divided into two categories: (a) early response genes that are transcribed in response to

growth factors, such as C-JUN, C-FUS, MYC, and I-AP genes; (b) delayed response genes that are transcribed after early response genes, such as cyclins and CDK genes. Anti-mitogens can also inhibit the initiation of cell division by preventing the expression and accumulation of CDKs [54–56]. In addition, anti-mitogens induce the expression and production of CKIs (Fig. 1). For example, TGFβ 2 is an important anti-mitogen, which increases the expression of p15INK4B [57,58].

Cyclin D is present throughout the G1 phase, and its expression increases in response to growth factors [59]. During this phase, CDK4 and CDK6 also activate a family of transcription factors called E2F [60]. Retinoblastoma protein (RB) is capable of inhibiting E2F [61]. During this phase, CDK4 and CDK6 induce the activation of E2F by inhibition of RB protein (Fig. 1) [62]. E2F transcription factors initiate the expression of DNA synthesis protein genes: cyclins A, B, and E [51,63]. RB can be phosphorylated by CDKs at several sites, thereby reducing the affinity of RB to bind to E2F [64]. As a result, RB disassociation from E2F causes to abolish the RB inhibitory effect on E2F (Fig. 1) [65]. E2F primarily cause to express essential genes for replication and essential genes for the S phase, and secondly cause to cyclin A and E genes expression, which in later stages increase E2Fs activity through RB phosphorylation [63]. Moreover, E2F can increase its own expression through a self-regulatory cycle (positive feedback) [66]. These events lead to simultaneous increases in E2F and CDK2 at the end of G1 phase, which facilitates the initiation of S phase (Fig. 1) [67]. Similarly, increased CDK2 during the S phase and increased CDK1 during the G2 phase both lead to the phosphorylation of RB protein during S, G2, and early M phases [68]. When the cell completes the anaphase and enters the early phase of the cell cycle (G0 or G1), the activity of CDKs decreases, which results in the dephosphorylation of RB, thus RB regains its ability to inhibit E2F activity. As a result, the cell does not undergo division again until it receives the next signal from growth factors [69,70].

To initiate the S phase, CDK2, CDK4, and CDK6 must inhibit the degradation mechanisms of cyclins A and E (Figs. 1 and 2) [71,72]. On the other hand, p27 (CKI27) is also inhibited during the S phase. p27 prevents CDK2 activation during the G1 phase. CDK2, CDK4, and CDK6 carry out p27 phosphorylation at several sites, resulting in p27 inhibition [73,74]. This phosphorylation causes p27 to bind to SCF (protein ubiquitin kinase) and consequently to be degraded in proteasomes [75]. The reason that several p27 sites need to be phosphorylated, is that p27 is only a weak substrate for CDK2, CDK4, and CDK6. Due to the weak substrate property of p27, it is only degraded when it is phosphorylated at several sites. This makes the process highly sensitive, so when all the essential sites are not phosphorylated because CDK activity is low, p27 remains active, but when the CDK activity is high p27 is degraded [76]. Surprisingly, p27 can also inhibit CDK2, CDK4, and CDK6. Thus p27 can take part in a negative feedback loop. In addition, mitogens can also stimulate and activate the kinases, which can phosphorylate p27 leading to its inactivation in similar ways [77]. During the S phase, CDK2 eventually leads to DNA replication by protein phosphorylation (Fig. 2) [78].

The M phase is induced by formation of a complex between CDK1 and cyclins A, B (Fig. 3) [79]. Since CDK1 is present throughout the cell cycle, the levels of cyclin A, B that accumulate at the end of S phase govern the complex formation [80]. The CDK1-CyclinA/B complex activity begins when DNA replication is complete [81]. This complex carries out protein phosphorylation, allowing the chromosomes to become more dense, the nuclear envelope to disappear, and the mitotic spindle to form, altogether necessary for mitosis to occur [82]. The activity of this complex is reduced during the occurrence of the mitotic event until it is completely inactivated during the anaphase [83]. This complex is phosphorylated by WEE1 at both tyrosine 15 and threonine 14 sites leading to its inactivation. However, it is dephosphorylated by CDC25C at both sites leading to its activation (Fig. 3). In cells with WEE1 deficiency, CDK1 is activated at an earlier stage and premature mitosis can occur. A defect in CDC25C is also equivalent to cell cycle arrest in the G2 phase [84]. Thus activation of CDK1 is due to rapid activation of CDC25C and inactivation of WEE1. Rapid activation of CDK1 occurs when WEE1 is inactivated and CDC25C is activated [85]. This pathway is initiated at the end of DNA replication, during which activated CDC25C dephosphorylates and activates the CDK 1-cyclinA complex. This activated CDK simultaneously activates more CDC25C and inactivates more WEE1 (Fig. 3) [86]. CDK1 stimulates chromatid condensation, chromatid binding to the mitotic spindle, and at the same time causes the disappearance of the nuclear membrane to provide the conditions for the separation of the chromosomes after replication [87].

5. CDK1 mechanism of actions

The nuclear membrane is double-layered, in a similar manner to the rough endoplasmic reticulum, and has pores called porins. The inner membrane is covered with lamin intermediate filaments [88]. These filaments play a role in maintaining the nuclear membrane structure and are composed of three types of lamin A, B, C. CDK1 phosphorylates the same serine residues in all three types of lamins. This phosphorylation results in the depolymerization of these filaments. Depolymerization of the lamins causes the membrane support structure to be lost, resulting in the membrane being rapidly destroyed [89,90]. Furthermore, CDK1 also phosphorylates and breaks down the nuclear pores [91]. In addition, phosphorylation of nuclear membrane proteins weakens the association between chromatin and the membrane, which taken together result in the disappearance of the membrane [91,92]. Moreover CDK1 induces the initiation of mitotic spindle production [93]. The mitotic spindle is composed of microtubules, and binds to the chromosomes in association with kinetochore proteins. This microtubule network is synthesized by centrosomes. The centrosomes themselves are made up of γ-tubulin units. Together with the mitotic spindle, the centrosomes cause the chromosomes to become separated. This separation results in the transfer of each sister chromatid to a daughter cell. The key function of CDK1 in this process is to induce mitotic spindle formation and activate the production of primitive nuclei [94–96].

As mentioned above, each sister chromatid is attached to microtubules by the kinetochore. The sister chromatids are not separated because they are interconnected in the centrosomal region by the cohesin complex. During this process a protease called separase, results in cleavage of cohesin, providing the conditions for chromatid separation [97,98]. Separase is inhibited by the securin protein, and securin is also ubiquitinated by APC/C, which results in the proteasomal degradation of securin [99]. APC/C is also activated by CDK1 by the phosphorylation of multiple sites on APC/C. The APC/C activation process is also controlled by checkpoints to ensure proper attachment of the chromatids to the mitotic spindle (Fig. 3) [100–102].

6. Inhibition of CDK1 induces exit from mitosis

The process of mitosis is largely governed by the phosphorylation of substrates by CDK1 [82,103]. These substrates must be dephosphorylated in order to exit mitosis [104]. This dephosphorylation occurs when CDK1 is inhibited or inactivated. Inactivation of CDK1 is carried out by APC/C-CDC20-dependent proteasomal degradation [105], therefore CDK1 can cause its own degradation by activating APC/C [106]. In addition to deactivating CDK1, dephosphorylation of substrates by phosphatases is required for mitosis exit. This is carried out by phosphatase 1, 2 A proteins (Fig. 3) [107].

7. Untimely CDK1 inhibition due to WEE1 activation arrests mitosis

DNA damage checkpoints can detect DNA damage, in order to stop the cell cycle and activate repair pathways [108]. These mechanisms are different depending on the severity of the DNA damage [109]. The most severe type of damage is the double strand break (DSB). If the M phase begins, this damage will result in the loss of part of the DNA [110]. Other

Table 1
WEE1 inhibitors.

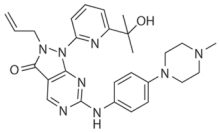
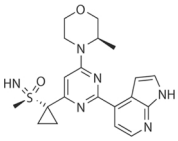
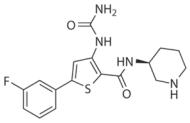
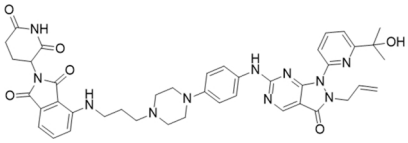
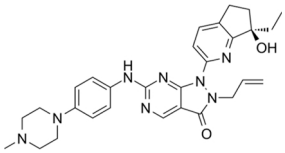
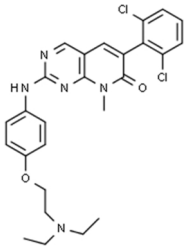
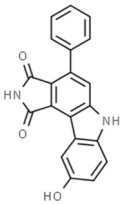
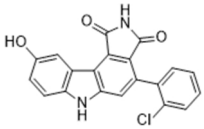
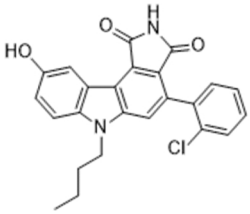
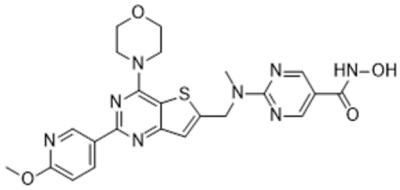
Inhibitor	Molecular Weight	Chemical Formula	Chemical Name	Chemical Structure	Refs
AZD1775 (MK1775)	500.6	$C_{27}H_{32}N_8O_2$	2-Allyl-1-(6-(2-hydroxypropan-2-yl) pyridin-2-yl)– 6-((4-(4-methylpiperazin-1-yl)phenyl) amino)– 1 H-pyrazolo[3,4-*d*]pyrimidin-3(2 H)-one		[150]
AZD6738 (Ceralasertib)	412.51	$C_{20}H_{24}N_6O_2S$	4-{4-[(3 R)– 3-Methyl-4-morpholinyl]– 6-[(2 R)– 2-(S-methylsulfonimidoyl)cyclopropyl]– 2-pyrimidinyl]– 1 H-pyrrolo[2,3-*b*]pyridine		[151]
AZD7762	362.42	$C_{17}H_{19}FN_4O_2S$	3-(Carbamoylamino)– 5-(3-fluorophenyl)-N-[(3 S)– 3-piperidinyl]– 2-thiophenecarboxamide		[152]
ZNL-02–096	799.893	$C_{42}H_{45}N_{11}O_6$	4- ((3- (4- (4- ((2-Allyl-1- (6- (2-hydroxypropan-2-yl) pyridin-2-yl)– 3-oxo-2,3-dihydro-1 H-pyrazolo [3,4-d] pyrimidin-6-yl)amino)phenyl)piperazin-1-yl)propyl) amino)– 2- (2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione		[153]
ZN-c3	526.63	$C_{29}H_{34}N_8O_2$	(R)– 2-allyl-1-(7-ethyl-7-hydroxy-6,7-dihydro-5 H-cyclopenta[*b*]pyridin-2-yl)– 6-((4-(4-methylpiperazin-1-yl)phenyl)amino)– 1,2-dihydro-3 H-pyrazolo[3,4-*d*] pyrimidin-3-one		[154]
PD166285	512.431	$C_{26}H_{27}Cl_2N_5O_2$	6-(2,6-Dichlorophenyl)– 2-([4-[2-(diethylamino) ethoxy]phenyl]amino)– 8-methylpyrido[2,3-*d*] pyrimidin-7(8 H)-one		[155]
PD0407824	328.32	$C_{20}H_{12}N_2O_3$	9-Hydroxy-4-phenylpyrrolo[3,4-*c*]carbazole-1,3 (2 H,6 H)-dione		[156]
WEE1 Inhibitor (i)	362.76	$C_{20}H_{11}ClN_2O_3$	4-(2-Chlorophenyl)– 9-hydroxypyrrolo[3,4-*c*] carbazole-1,3-(2 H,6 H)-dione		[157]
WEE1 Inhibitor (ii)	418.87	$C_{24}H_{19}ClN_2O_3$	6-Butyl-4-(2-chlorophenyl)– 9-hydroxypyrrolo[3,4-*c*] carbazole-1,3-(2 H,6 H)-dione		[158]
CUDC-907 (Fimepinostat)	508.55	$C_{23}H_{24}N_8O_4S$	N-hydroxy-2-(((2-(6-methoxypyridin-3-yl)– 4-morpholinothieno[3,2-*d*]pyrimidin-6-yl)methyl) (methyl)amino)pyrimidine-5-carboxamide		[159]

Table 2
Current studies of WEE1 inhibition for cancer therapy.

Brief title	Type of study	Sample population	Sample markers	Drugs/ intervention	Brief results	Brief methods	Reference
Combination of WEE1 inhibitor + cisplatin to increase DNA damage and apoptosis in gastric cancer.	in vitro *in vivo*	in vitro; MKN45, N87, AGS, MGC803, SNU1, KATOIII, HGC27, SNU5 cell lines in vivo; MGC803 xenograft 6 week old female nude mice	Gastric adenocarcinoma, myc+ , erbB2 + , DDC +	WEE1 inhibitor (AZD1775), cisplatin	AZD1775 in combination with cisplatin inhibited growth in GC cells. Induced DNA damage, apoptosis and G2-M arrest. Reduced migration and invasion of GC cells. AZD plus cisplatin reduced tumor growth in vivo.	Cell viability assay, cell apoptosis assay, cell cycle assay, immunofluorescence staining, immunoblotting, in vivo study (xenograft mice), immunohistochemistry, statistical analysis	[134]
Effect of WEE1 kinase and ATR co-operation via DDR pathway on AML	in vitro	in vitro; THP-1, U937, MOLM13 cell lines	histiocytic lymphoma, AML-M5, p53 + , FLT3-ITD+ , RAD51 + , CHK1 + , BRCA1 +	WEE1 inhibitor (AZD1775), ATR inhibitor (VE-822)	Inhibition of ATR led to decreased proliferation and induction of apoptosis in AML cells. Inhibition of WEE1 led to increased ATR inhibition. Simultaneous inhibition of both WEE1 and ATR pathways led to defects in mitochondrial membrane potential. Simultaneous inhibition resulted in cell cycle arrest and DNA damage.	MTT assay, lactate dehydrogenase release assay, comet assay, Western blotting, cell cycle progression assay, chromatin fraction assay	[135]
Activity mechanism of CUDC-907 in AML	*In vitro* in vivo	in vitro; THP-1, CMS, OCIAML3, MOLM13, U937, MV4–11, CTS cell lines, AML patient samples [76] in vivo; MV4–11 xenograft NGS-SGM3 mice	in vitro*:* AML-M5, AML-M1, AML-M7, FLT3-ITD+ , MLL AF9 + , CAML AF10 + , MLL AF6 + , MLL AF4 + , MLL AF9 + , NPM1 + patients: AML-M1, AML-M2, AML-M3, AML M4, AML-M5, NPM1 + , FLT3-ITD+ , DUP MLL+ , CEBPA+ , DEGCAN+ , AML1-ETO+ , PML-RAR+ , IDH2 + , DNMT3A+ , HOX11 + , MLL AF6 + , IDH1 + , CBFB-MYT1 + , CKIT+ , NRAS+ , SF3B1, TET2, GATA2	CUDC-907 (fimepinostat)	CUDC-907 induced apoptosis in AML cells through inhibition of histone deacetylases and PI3K, which induced a decrease in WEE1, CHK and RRM1, and an increase in Bim producing more DNA damage. The C-MYC gene was down-regulated byPI3K inhibition.	MTT assay, Western blotting, AnnexinV-FITC staining, shRNA knockdown of Bim, ectopic overexpression of MCL-1, BCL-XL and C-MYC, alkaline comet assay, chromatin fraction assay, RTQ-PCR, DNAPK assay	[136]
Pancreatic cancer cells are susceptible to chemo radiation through defects in the HR pathway due to the inhibition of WEE1 kinase.	*In vitro* in vivo	in vitro; MIAPaCa-2, PANC-1, DLD-1 cell lines in vivo; patient derived tumor xenograft in mice	PaCa: myc+ , myb+ , ras+ , fos+ , sis+ , p53 + , abl-, ros-, src-	WEE1 inhibitor (AZD1775), Gemcitabine	Inhibition of WEE1 in different cell lines showed a different sensitizing effect. Treatment with AZD1775 increased YH2AX, and subsequently more DNA DSBs. Depending on the HR pathway activity, PaCa cells showed a different sensitivity to chemoradiation. AZD1775 can sensitize patient-derived tumor xenografts to gemcitabine/radiation. AZD1775 delayed tumor volume doubling.	Clonogenic survival assay, flow cytometry, immunoblotting, immunofluorescence, tumor growth study, statistical analysis	[139]
Effects of combined WEE1 and ATR inhibitors on tumor growth in TNBC	*In vitro*	in vitro; MDA-MB-231, HS 5778 T, MDA-MB-157, BT-549, HCC1937, HCC70, MCF7, MCF 10 A cell lines	Cell lines derived from fibrocystic disease or carcinoma. WNT7B+ , her2/neu+ , p53 + , BRCA1 +	WEE1 inhibitor (AZD1775), AZD6738, BRCA1 siRNA	AZD1775 and AZD6738 have anti-proliferative effect on TNBC. WEE1 inhibition plus ATR inhibition increased DNA damage and replication stress in TNBC cells. Mitotic catastrophe was induced in cells with early	Colony formation assay, siRNA and lentivirus transfection, Western blotting, immunofluorescence, flow cytometry, xenograft experiments, TUNEL staining of tissue	[138]

(*continued on next page*)

Table 2 (*continued*)

Brief title	Type of study	Sample population	Sample markers	Drugs/ intervention	Brief results	Brief methods	Reference
					mitosis The combination of WEE1 and ATR inhibition sensitized TNBC cells to cisplatin, and showed a synergistic antitumor effect in xenograft mouse model.		
Effects of WEE1 inhibitors in combination with radiotherapy in cervical cancer.	*In vitro* in vivo	in vitro; SiHa, HeLa cell lines in vivo; CX6M6 and CX21M6 xenograft mice	Squamous cell carcinoma, adenocarcinoma, p53 + , pRB+	WEE1 inhibitor (AZD1775), Radiation	AZD1775 in combination with radiotherapy caused a significant decrease in colony formation in cervical cancer lines. AZD1775 plus RT significantly increased apoptosis in both cell lines. Increased DNA damage and G2-M checkpoint abrogation. AZD1775 in combination with radiation substantially delayed tumor growth in xenograft mice	Western blotting, MTT assay, colony forming assay, ELISA, apoptosis assay, cell cycle analysis, immunohistochemistry, TUNEL assay, patient derived xenograft in vivo model, statistical analysis	[137]
Simultaneous inhibition of WEE1 and CHK1 can induce an anticancer effect in melanoma.	In *vitro* in vivo	in vitro; WM239, WM45.1, WM983B, WM1366 cell lines in vivo; 3-FEMXI xenograft mice	Melanoma	WEE1 inhibitor (AZD1775), AZD7762, WEE1 siRNA	Combination of AZD1775 and AZD7762 had synergistic effects on melanoma cells. Simultaneous inhibition of WEE1 and CHK1, 2 led to decreased survival and irreversible inhibition of melanoma growth. Inhibition of WEE1 and CHK1, 2 led to DNA damage and apoptosis. AZD1775 and AZD7762 induced premature mitosis in melanoma.	siRNA transfection, viability assay, CalcuSyn analysis, Western blotting, flow cytometry, in vivo xenograft mouse study, statistical analysis	[140]
Increased sensitivity of resistant non-small cell lung cancer (NSCLC) cells to chemotherapy after treatment with WEE1 inhibitors	In vitro	in vitro; HL299, PC9/G2 cell lines Histological specimens from NSCLC patients	EFGR-mut, EFGR T790-mut	WEE1 siRNA, anti-epidermal growth factor receptor (EGFR), TKIs, cisplatin (DDP), gemcitabine (GEM)	WEE1 had high expression in resistant NSCLC cell lines. Af conventional interventions in NSCLC cells, WEE1 expression is increased to a greater extent. Inhibition of WEE1 increases the sensitivity of the resistant NSCLC cells to conventional therapies.	Immunohistochemistry assay, RNA interference, western blot analysis, Q-PCR, MTT assay, colony formation assay, statically analysis	[141]

types of damage include, single strand break (SSB) and changes in the nucleotide structure (oxidative damage) [111]. Detection of DNA damage is performed by two major ATM and ATR sensors, which are both protein kinases (Fig. 3) [112]. ATM is specifically activated after double-strand breaks, while ATR is activated in response to a wide variety of damage, but it is known to recognize the presence of single stranded DNA resulting from SSBs [112–114]. ATR and ATM bind to the damaged DNA and activate adaptor proteins, as well as two protein kinases, CHK1 and CHK2. The adaptor protein recruits and activates CHK1 and CHK2 [115]. CHK1 and CHK2 activate repair mechanisms and pause the cell cycle until the DNA is repaired [116]. CHK1 and CHK2 cause cell cycle arrest due to phosphorylation and activation of WEE1 [117]. Activation of WEE1, which coincides with inhibition of CDC25C, inhibits CDK1 during the G2 phase (Fig. 3) [118,119]. The ATM and ATR will continue activating WEE1 until the DNA replication is complete. This process makes the inhibition of M phase dependent on the completion of G2 phase [120].

8. WEE1 inhibition results in apoptosis

Cell cycle control systems and checkpoints work to prevent cells with mutations from developing malignant transformation and cancer. Cells that have DNA damage are stopped from proliferating before DNA replication or chromatid separation occurs [121]. Under these conditions if damage cannot be repaired, the apoptosis pathway is activated. However, in cells where WEE1 is activated, and if the cells cannot repair the damage, the apoptosis pathway cannot be activated, and the cells accumulate rather than die, leading to possible development of cancer [122,123]. Indeed, these damaged cells survive because the cell cycle is arrested, but they are not capable of activating apoptosis [124]. However, initiation of the M phase will lead to cell death in those cells that have damaged DNA and cannot repair the damage. This cell death is due to mitotic catastrophe [125,126]. Mitotic catastrophe occurs due to premature and inappropriate mitosis [127]. Mitotic catastrophe is a different process to apoptosis, because apoptotic pathways are not

activated, neither is the mitochondrial membrane permeablized. The main marker of mitotic catastrophic is the micronuclei that result from double strand breaks in DNA [128,129].

As mentioned, WEE1 kinase is a key regulator of the G2-M checkpoint which prevents initiations of M phase by inhibiting CDK1. Therefore when cells are damaged, WEE1 kinase leads to cell cycle arrest at the G2-M stage (Fig. 3) [84,117]. Inhibition of WEE1 prevents the G2-M arrest resulting in premature mitosis. Mitotic catastrophe and apoptosis are the result of this premature mitosis because the DNA damage is not repaired [118,125]. Therefore, it has been proposed that inhibition of WEE1 could be a therapeutic strategy in the treatment of cancer.

9. Targeting WEE1 as a cancer therapy

Recent studies have shown that WEE1 inhibition can overcome chemotherapy resistance in the treatment of cancer cells [130,131]. This is because many types of chemotherapy are based on producing DNA damage in cancer cells, but if WEE1 protein kinase is activated it will prevent the death of cells with DNA damage. Thus, inhibition of WEE1 could be a new therapeutic strategy in the treatment of cancer [132,133]. Several studies have been conducted to evaluate the efficacy of WEE1 inhibition in the treatment of cancer.

In one study, human gastric cancer (GC) cell lines including, MKN45, N87, AGS, MGC803, SNU1, SNU5, KATO3, HGC27, SNU5, were used to demonstrate the effects of a small molecule WEE1 inhibitor called AZD1775 (Table 1), in combination with cisplatin to treat gastric cancer. AZD1775 or Adavosertib was designed to be a specific WEE1 inhibitor. It has been shown that WEE1 inhibition alone was capable of inhibiting growth, invasion and migration in human GC cells. In GC cell lines in vitro and in xenograft models of human GC in vivo, it has been shown that inhibition of WEE1 in combination with cisplatin exerted anticancer effects by increasing DNA damage. The mechanism of WEE1 inhibition against GC, involves dephosphorylation and activation of CDK1. This will cause abrogation of the G2-M checkpoint. Due to the replication stress caused by G2-M checkpoint abrogation and homologous recombination (HR) failure (due to the role of WEE1 in HR) it leads to increased DNA damage and induction of apoptosis. The data from this study showed that GC cells were less capable of invasion and migration when exposed to the combination of AZD1775 plus cisplatin. Taken together, the data from this study suggest a promising therapeutic approach for the inhibition of WEE1 in combination with cisplatin in gastric cancer (Table 2) [134].

Another study used acute myeloid leukemia (AML) cell lines including, THP-1 and MOLM13. Over-expression of CHK1 and WEE1 in these cell lines has been confirmed. These cell lines were treated with two different inhibitors, AZD1775 and VE-822 (Table 1). VE-822 is a close analogue of VE-821, which is an inhibitor of ATM and ZTR. VE-822 has better solubility, and more specificity for ATR inhibition. In this study it was reported that VE-822 induced apoptosis and inhibited proliferation in AML cell lines. Different AML cell lines showed different sensitivity to VE-822. When AZD1775 was combined with VE-822 it synergistically reduced proliferation in AML cell lines, and disrupted mitochondrial activity. In addition, this combination resulted in downregulation of WEE1, less CDC25C phosphorylation (p-CDC25c), and inhibition of ribonucleotide reductase enzyme subunits (RRM1, RRM2). As a result, apoptosis was induced by replication stress. Also, these compounds (AZD1775, VE-822) increased DNA damage with more apoptosis in AML cell lines (Table 2) [135].

In another study, a variety of AML cell lines (MV4–11, THP1, U937, MOLLM13), as well as AML patient samples and mouse AML xenograft models were used. Pharmacological intervention with CUDC-907 (Table 1), designed as an inhibitor of PI3K and HDAC, was tested. CUDC-907 showed promising results in a variety of cancers, and is currently undergoing phase I and II clinical trials to treat lymphoma, multiple myeloma, and relapsed progressive solid tumors. In this study, it was shown that CUDC-907 could reduce the expression of WEE1, CHK1, and the catalytic subunit of ribonucleotide reductase (RRM2) in patient samples and cell lines. In addition, CUDC-907 induced apoptosis by affecting three different factors: MCL-1, BIM, and C-MYC. Taken together, these events suggest the induction of replication stress and DNA damage. In vivo studies on xenograft models showed the efficacy of CUDC-907 along with low toxicity and few side effects for AML treatment (Table 2) [136].

In a study designed to investigate the effect of WEE1 inhibitors in combination with radiation therapy (RT) against cervical cancer, it was shown that inhibition of WEE1 by AZD1775 could effectively enhance the effects of RT. In a cell line-derived xenograft model and a cervical cancer patient-derived xenograft model (PDX), combination treatment with AZD1775 plus RT significantly delayed tumor growth. It was also observed that the WEE1 inhibitor had no serious side effects in mice, and AZD1775 may have appropriate systemic tolerability for in vivo use. In this study, inhibition of WEE1 by AZD1775 was reported to abrogate the G2-M checkpoint (Table 2) [137].

In a study designed to show the effects of WEE1 inhibition in combination with ATR inhibition in the treatment of triple negative breast cancer (TNBC), it was shown that the combination of AZD1775 and the ATR inhibitor AZD6738 (Table 1), had significant antitumor effects against TNBC. Significant decrease in cell survival was observed following the combination treatment. The study also demonstrated that the combination of AZD1775 and AZD6738 further increased the ability of cisplatin to induce cell death. Finally, a synergistic effect was observed between WEE1 inhibition and ATR inhibition as a future TNBC treatment (Table 2) [138].

Another study showed that inhibition of WEE1 increased the sensitivity of pancreatic cancer cells to gemcitabine and RT (chemoradiation therapy). Although WEE1 inhibition has been shown to cause G2-M checkpoint abrogation, the susceptibility of pancreatic cancer cells does not appear to be due to this mechanism. The proposed hypothesis was that the sensitivity to chemoradiation was due to a defect in the HR pathway. They showed that treatment with AZD1775 caused higher levels of YH2AX (H2A histone family member X) due to a defect in the HR pathway. An increase in YH2AX levels would be translated into more DNA DSBs and also increased the sensitivity to chemoradiation. In this regard, it has been observed that cells with an intact HR pathway are more sensitive to WEE1 inhibition, but cells that already have a defect in the HR pathway (although intrinsically more sensitive to chemoradiation) do not further respond to WEE1 inhibition (Table 2) [139].

One study investigated the role of a combination of a WEE1 inhibitor with a CHK1 inhibitor in a panel of melanoma cell lines as an anticancer approach. They tested AZD1775 and AZD7762 as a CHK1 inhibitor (Table 1). It was shown that the use of both inhibitors combined had potent antitumor effects, both in vitro and in xenograft models, whilst neither showed much effect as a single agent. In addition, this combination reduced cell survival and increased premature mitosis due to increased replication stress and DNA damage in melanoma cell lines. Likewise, cell death occurred in melanoma xenografts in mice treated with AZD1775 and AZD7762. Overall, the inhibition of WEE1 with AZD1775 (or with siRNA) can lead to replication stress and loss of genome integrity due to nucleotide starvation, and increased DNA damage (Table 2) [140].

Another study was performed to understand the association between WEE1 expression and chemotherapy resistance in non-small cell lung cancer (NSCLC). It was observed that WEE1 inhibition sensitized resistant NSCLC cell lines to chemotherapy. In addition, inhibition of WEE1 kinase could enhance the efficacy of other types of NSCLC therapies, such as tyrosine kinase inhibitors or anti-EGFR antibodies. In clinical studies, it was found that 58% of patients had increased expression of WEE1 after chemotherapy treatment, which could be counteracted by WEE1 inhibition. Moreover, chemotherapy doses could be reduced by 2–3 times after WEE1 inhibition because the sensitivity to chemotherapy drugs was increased. Overall, this study suggested that WEE1

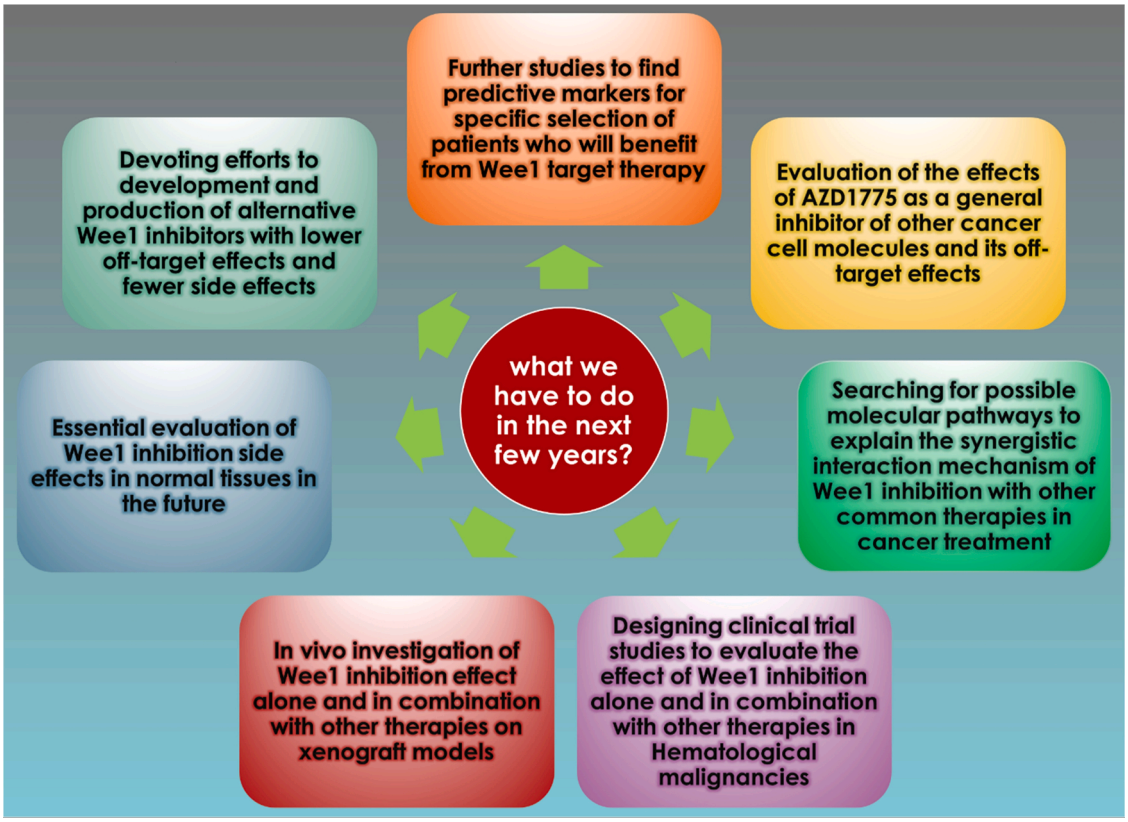


Fig. 4. Topics for further investigation and recommendations.

kinase could be a target for the treatment of refractory NSCLC (Table 2) [141].

Bifunctional degraders have also been used to increase the specificity and sensitivity of WEE1 inhibition. Degraders have two functional parts, one that is specific and binds to the target protein, and the other that contains the cereblon-binding ligand (pomalidomide), which uses E3 ubiquitin ligase to degrade the target protein. ZNL-02–096 as a bifunctional degrader compose of two protein binding ligands: AZD1775 that bind to WEE1 and pomalidomide which facilitates WEE1 degradation in the ubiquitin ligase complex process [142,143]. In a study using ZNL-02–096 to increase the sensitivity of AZD1775, it was shown that ZNL-02–096 inhibited WEE1 by 10 times lower dose than AZD1775. Also, while AZD1775 inhibits other cell cycle kinases such as PLK1 (Polo Like Kinase 1), ZNL-02–096 has been shown to more specifically degrade only WEE1. In addition, ZNL-02–096 has been used to increase the efficacy of olaparib in the treatment of ovarian cancer cells. The results indicated that ZNL-02–096 increased DNA damage induced by olaparib by degrading overexpressed WEE1 and thus inhibiting G2-M checkpoint thereby eliminating cell cycle arrest, which eventually led to ovarian cancer cells apoptosis. ZNL-02–096 also by degrading WEE1, which is dependent on ubiquitin ligase-related processes, synergistically enhances the anti-tumor and anti-proliferative effects of olaparib in ovarian cancer [144].

ZN-c3 is onother selective WEE1 kinase inhibitor which exhibit potent anti-proliferative activity across multiple cancer cell lines. ZN-c3 is under various ongoing clinical trials which investigating WEE1 kinase inhibition utility in different conditions such as ovarian cancer, solid tumors, osteosarcoma, etc (Clinicaltrials.gov). Although ZN-c3 is more specific than AZD1775, its inhibitory function has been observed on several different kinases. In studies, the anti-proliferative and anti-tumor properties of ZN-c3 have been observed in various cancer cell lines and human xenograft tumors, respectively. In addition, ZN-c3 causes tumor regression in clinical trials in ovarian cancer and non-small cell lung cancer. One of the advantages of ZN-c3 is its high maximum tolerated dose, so at doses much lower than AZD1775 it can exert an equal amount of growth/proliferation inhibition. However, side effects such as fatigue, vomiting, diarrhea, and nausea have been seen in most cases in the studies. Overall, ZN-c3 has shown its effectiveness in the treatment of cancer cells as a single treatment with high tolerance and appropriate safety [145,146].

Another interesting compound in this area is rhodium (iii), which has been suggested as an inhibitor of WEE1 both in vitro and in vivo. This compound is also known as a metal-based inhibitor, which increases DNA damage and apoptosis in triple negative breast cancer cells by inhibiting WEE1 kinase. That is why in cases of patient recurrence or resistance to treatment, these metal based drugs are used to get the most therapeutic benefits. Rhodium alone has been shown to reduce proliferation in breast cancer cells while leaving normal cells intact, indicating that this compound has selectivity only to cancer cells. This selectivity can be presumed as an advantage of this therapeutic strategy (inhibition of WEE1) over conventional chemotherapy, that also affects normal cells. The efficacy of rhodium (iii) in various breast cancer cell types is also inversely related to WEE1 expression, indicating that WEE1 overexpression is directly related to tumor recurrence and progression, which can be overcome by inhibiting WEE1 expression. But rhodium (iii) only reduces WEE1 function, and WEE1 mRNA expression continues, leading to a return to WEE1 function in the long run. siRNA, which is a WEE1 inhibitor, can be used to correct these defects of rhodium (iii) in order to prevent the return of WEE1 function by inhibiting its mRNA [147–149].

10. Discussion

In this review article we have presented some new insights into how cell cycle checkpoints play a role in the cell cycle, and suggested the potential of checkpoint inhibitors and activators as therapeutic targets

for treating a variety of different cancers. WEE1 is a critical target for inhibitors, which can enhance the efficacy of other types of conventional therapy and overcome the resistance to conventional treatment. WEE1, which primarily functions by inhibiting the CDK1-cyclinB complex, also affects the function of many other proteins because of this activity. In general, the mechanism responsible for the anticancer effect of WEE1 inhibition is through the over-production of replication stress, DNA damage, premature mitosis, and ultimately cell death due to mitotic catastrophe. Studies have hypothesized that the action of WEE1 inhibition in causing DNA damage is multi-fold, for instance, reducing the binding of RPA to DNA, stalling the replication fork, or reducing the total cellular DNTP pool. Therefore, in order to better determine the DNA damage mechanism of WEE1 inhibition, the interaction of this protein with other molecules such as CDK2, CDK4, CDK6, CAK, CKI, cyclins D, E, APC/C-CDC20, and RR should be investigated more deeply in future studies (Fig. 4).

In most reports, the molecule AZD1775 was used as a WEE1 inhibitor, but because of its inhibitory activity against eight additional kinase enzymes, it has poor specificity for WEE1 inhibition. AZD1775 also has an intrinsic anticancer effect, which is independent of its WEE1 inhibition effect. Thus the non-specific activity of AZD1775 causes us to doubt any assumption made about the specific role of WEE1. Moreover, specific siRNAs used against WEE1 are unable to completely knockout its activity. This weakness is a serious problem in explaining the anticancer function of WEE1 inhibition, determining the physiological role of WEE1, and validating WEE1 inhibitors as a treatment. Therefore, the design and development of new more specific WEE1 inhibitors should be urgently considered (Fig. 4).

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

Acknowledgements

The research protocol was approved & supported by Student Research Committee, Tabriz University of Medical Sciences, Tabriz, Iran (registration code: 64850).

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